

Software requirement specification

For Online quiz system

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Sept 6 2026

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Revision History

Name	Date	Reason for change	Version
Nakshathira	06/09/26	Initial SRS document	1.0

Introduction

Purpose

The purpose of the Online Quiz System is to provide a simple and interactive platform for conducting multiple-choice quizzes. The system enables users to attempt quizzes, receive scores and instant feedback, and view their performance through leaderboards.

Intended Audience

This document is intended for the project team, developers, testers, evaluators, and users of the Online Quiz System. It defines the system's purpose, scope, features, interfaces, and requirements.

Product Scope

The Online Quiz System allows users to select quiz categories, attempt timed multiple-choice quizzes, receive instant feedback, view their scores, and compare performance through leaderboards. The system aims to make quiz participation simple, organized, and engaging.

References

1. JACKFRUIT – Software Engineering Project Guidelines 2026, Department of Computer Science and Engineering, PES University.

2. Overall Description

2.1 Product Perspective

The Online Quiz System is a standalone web-based application designed to provide an organized platform for conducting multiple-choice quizzes. It allows users to select a category, attempt a timed quiz, receive feedback, and view their scores and leaderboard rankings.

2.2 Product Functions

The system will provide the following major functions:

- Allow users to select a quiz category.
- Display multiple-choice questions.
- Conduct quizzes within a specified time limit.
- Record user responses.
- Calculate and display scores.
- Provide instant feedback.

- Maintain and display a leaderboard.

2.3 User Classes and Characteristics

The system will have two main user classes:

- User: Selects categories, attempts quizzes, receives feedback, views scores, and checks the leaderboard.
- Administrator: Manages quiz questions, categories, and quiz content.

2.4 Operating Environment

The system will operate as a web application and will be accessible through modern web browsers on desktop and mobile devices. The system will require an internet connection and a server-side application with a database for storing user, quiz, question, and result data.

2.5 Design and Implementation Constraints

The system must provide secure handling of user and quiz data. The application should use a structured database for storing and retrieving quiz information. The system must support timed quizzes and provide accurate scoring and leaderboard results.

2.6 Assumptions and Dependencies

It is assumed that users have access to a compatible web browser and a stable internet connection. The system depends on the availability of the application server and database. Quiz questions and categories are assumed to be maintained by the administrator.

3. External Interface Requirements

3.1 User Interfaces

The Online Quiz System will provide a simple and user-friendly web interface. The main interfaces will include:

- Login/Registration screen for user authentication.
- Home screen for accessing quiz categories.
- Quiz screen for displaying questions, answer options, and the remaining time.
- Result screen for displaying the score and feedback.
- Leaderboard screen for displaying user rankings.
- Administrative interface for managing quiz questions and categories.

The interface will provide clear navigation, readable content, consistent button placement, and suitable messages for invalid inputs or completed actions.

3.2 Software Interfaces

The system will interact with the following software components:

- Web browser: Provides access to the user interface.
- Application server: Processes user requests, quiz logic, scoring, and result processing.
- Database: Stores user information, quiz categories, questions, answers, scores, and leaderboard data.

Data will be exchanged between the user interface, application server, and database to support quiz operations and user management.

3.3 Communications Interfaces

The system will use standard web communication protocols for communication between the client and server.

- HTTP/HTTPS will be used for client-server communication.
- HTTPS will be preferred to protect user credentials and other sensitive data during transmission.
- Communication between the application and database will be handled through the application server.

4. Analysis Models

The system will be represented using analysis models to describe its structure and interactions.

4.1 Use Case Diagram

The Use Case Diagram will represent the interactions between the main users and the Online Quiz System.

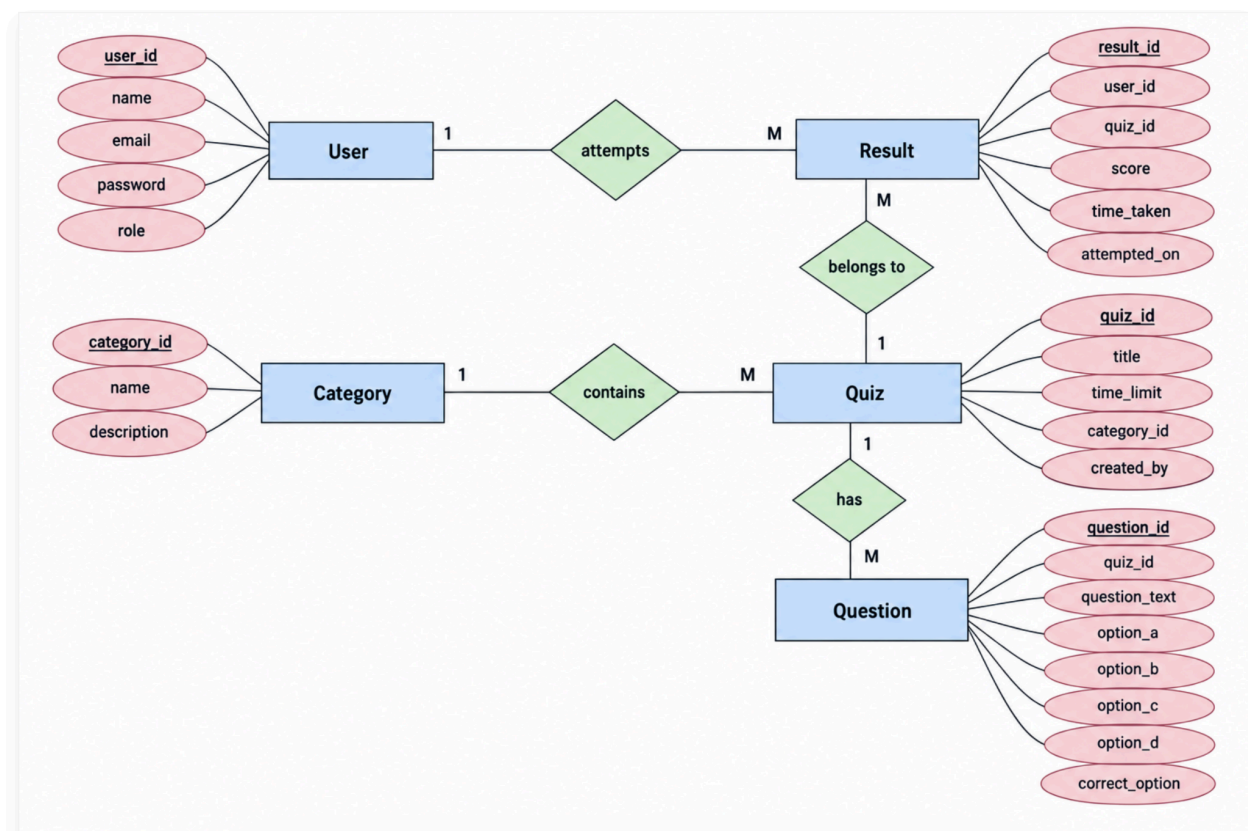
the primary actors are:

- User
- Administrator

Major use cases include:

- Register/Login
- Select Quiz Category
- Start Quiz
- Answer Questions
- Submit Quiz
- View Score
- View Feedback
- View Leaderboard
- Manage Questions
- Manage Categories

4.2 Entity-Relationship Diagram



5. System Features

5.1 Quiz Category Selection

5.1.1 Description and Priority

This feature allows users to select a quiz category before starting a quiz.

Priority: High

5.1.2 Stimulus/Response Sequences

1. The user opens the quiz system.
2. The system displays the available quiz categories.
3. The user selects a category.
4. The system displays the quizzes available under the selected category.

5.1.3 Functional Requirements

REQ-01: The system shall display the available quiz categories.

REQ-02: The system shall allow the user to select a quiz category.

REQ-03: The system shall display the quizzes associated with the selected category.

5.2 Timed Quiz and Question Attempt

5.2.1 Description and Priority

This feature allows users to attempt multiple-choice questions within a specified time limit.

Priority: High

5.2.2 Stimulus/Response Sequences

1. The user selects a quiz and starts it.
2. The system displays the quiz questions and starts the timer.
3. The user selects answers for the questions.
4. The system records the selected answers.
5. The system ends the quiz when the user submits it or the time limit expires.

5.2.3 Functional Requirements

REQ-04: The system shall display multiple-choice questions for the selected quiz.

REQ-05: The system shall provide multiple answer options for each question.

REQ-06: The system shall start a timer when the quiz begins.

REQ-07: The system shall record the user's selected answers.

REQ-08: The system shall automatically end the quiz when the time limit expires.

5.3 Scoring and Instant Feedback

5.3.1 Description and Priority

This feature evaluates the user's answers, calculates the score, and provides feedback after each response or on completion of the quiz.

Priority: High

5.3.2 Stimulus/Response Sequences

1. The user answers a question or completes the quiz.
2. The system compares the submitted answer with the correct answer.
3. The system provides appropriate feedback.
4. After quiz completion, the system calculates and displays the final score.

5.3.3 Functional Requirements

REQ-09: The system shall evaluate the user's submitted answers.

REQ-10: The system shall provide instant feedback on the user's answers.

REQ-11: The system shall calculate the user's score based on the number of correct answers.

REQ-12: The system shall display the final score after quiz completion.

5.4 Leaderboard and Performance

5.4.1 Description and Priority

This feature allows users to view their quiz performance and compare their scores through a leaderboard.

Priority: Medium

5.4.2 Stimulus/Response Sequences

1. The user completes a quiz.
2. The system stores the quiz result.
3. The user opens the leaderboard.
4. The system retrieves relevant scores and displays the rankings.

5.4.3 Functional Requirements

REQ-13: The system shall store the user's quiz score.

REQ-14: The system shall maintain leaderboard records for completed quizzes.

REQ-15: The system shall display users' rankings based on quiz performance.

5.5 Quiz Content Management

5.5.1 Description and Priority

This feature allows the administrator to manage quiz categories and questions used by the system.

Priority: Medium

5.5.2 Stimulus/Response Sequences

1. The administrator logs into the system.
2. The administrator selects quiz content management.
3. The administrator adds, updates, or removes questions or categories.
4. The system validates and stores the changes.

5.5.3 Functional Requirements

REQ-16: The system shall allow the administrator to add quiz questions.

REQ-17: The system shall allow the administrator to update quiz questions.

REQ-18: The system shall allow the administrator to delete quiz questions.

REQ-19: The system shall allow the administrator to manage quiz categories.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

REQ-20: The system shall provide quiz questions and user actions with minimal delay during normal operation.

REQ-21: The system shall record quiz responses and scores accurately without noticeable delay.

6.2 Safety Requirements

The system shall prevent loss of a user's submitted answers and quiz results during normal operation. The system shall handle expired quiz sessions without incorrectly accepting additional responses.

6.3 Security Requirements

REQ-22: The system shall require authentication for access to user accounts.

REQ-23: The system shall restrict administrative functions to authorized administrators.

REQ-24: The system shall protect user credentials and sensitive information during transmission.

6.4 Software Quality Attributes

The system shall be:

- Usable, with a clear and simple quiz interface.
- Reliable, with accurate scoring and result storage.
- Maintainable, with modular system components.
- Testable, with clearly defined and verifiable requirements.
- Responsive across commonly used desktop and mobile web browsers.

6.5 Business Rules

- A user must select a quiz before attempting its questions.
- A quiz must have a defined time limit.
- A user's score shall be calculated from the answers submitted during the quiz.
- Only authorized administrators may modify quiz questions and categories.
- Leaderboard rankings shall be based on recorded quiz performance.

7. Other Requirements

7.1 Database Requirements

The system shall maintain a structured database for storing user details, quiz categories, quizzes, questions, answers, and quiz results. The database shall support reliable storage and retrieval of information required for quiz operations and leaderboard generation.

7.2 Compatibility Requirements

The system shall be accessible through commonly used modern web browsers and shall provide a consistent user experience across desktop and mobile devices.

7.3 Data Management Requirements

The system shall maintain the accuracy and consistency of stored quiz and user data. Quiz results shall be retained so that user performance and leaderboard information can be displayed when required.

Appendix A: Glossary

SRS – Software Requirements Specification.

MCQ – Multiple-Choice Question.

UI – User Interface.

ER Diagram – Entity-Relationship Diagram used to represent entities and their relationships in the system.

User – A person who accesses the system to participate in quizzes.

Administrator – An authorized user responsible for managing quiz questions and categories.

Quiz – A collection of multiple-choice questions presented to a user within a specified time limit.

Leaderboard – A ranking of users based on their recorded quiz performance.

Quiz Result – The score and related information recorded after a user completes a quiz.

Appendix B: Field Layout

1. User

Field	Length	Data Type	Description	Is Mandatory
user_id	10	Integer	Unique user identifier	Y
name	100	String	Name of the user	Y
email	100	String	User email address	Y
password	255	String	Hashed user password	Y
role	20	String	User or Administrator role	Y

2. Category

Field	Length	Data Type	Description	Is Mandatory
category_id	10	Integer	Unique category identifier	Y
name	50	String	Name of the quiz category	Y
description	200	String	Description of the category	N

3. Quiz

Field	Length	Data Type	Description	Is Mandatory
quiz_id	10	Integer	Unique quiz identifier	Y
title	100	String	Title of the quiz	Y
time_limit	5	Integer	Time limit in minutes	Y
category_id	10	Integer	Category associated with the quiz	Y
created_by	10	Integer	Administrator who created the quiz	Y

4. Question

Field	Length	Data Type	Description	Is Mandatory
question_id	10	Integer	Unique question identifier	Y
quiz_id	10	Integer	Quiz associated with the question	Y
question_text	500	String	Text of the question	Y
option_a	200	String	First answer option	Y
option_b	200	String	Second answer option	Y
option_c	200	String	Third answer option	Y
option_d	200	String	Fourth answer option	Y
correct_option	1	String	Correct answer option (A, B, C or D)	Y

5. Result

Field	Length	Data Type	Description	Is Mandatory
result_id	10	Integer	Unique result identifier	Y
user_id	10	Integer	User who attempted the quiz	Y
quiz_id	10	Integer	Quiz that was attempted	Y
score	5	Integer	Score obtained by the user	Y
time_taken	10	Integer	Time taken in seconds	Y
attempted_on	20	DateTime	Date and time of the attempt	Y

Report Requirements

Report Name	Fields	Description
1. Quiz Result Report	User Name, Quiz Title, Category, Score, Time Taken, Attempt Date	Displays individual quiz attempt results.
2. Leaderboard Report	Rank, User Name, Quiz Title, Score, Attempt Date	Displays top users based on quiz performance.

Sample Report Requirements: Include the fields to be included in the report

Quiz Result Report

User Name
Quiz Title
Category
Score
Time Taken
Attempt Date

Leaderboard Report

Rank
User Name
Quiz Title
Score
Attempt Date

APPENDIX-C:Requirement Traceability Matrix

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
1	REQ-01	System shall display available quiz categories.	ERD	DD-01	category.py	TC-01	STC-01
2	REQ-02	System shall allow the user to select a category.	ERD	DD-01	category.py	TC-02	STC-01
3	REQ-03	System shall display quizzes for the selected category.	ERD	DD-02	quiz.py	TC-03	STC-02
4	REQ-04	System shall display multiple-choice questions.	ERD	DD-03	question.py	TC-04	STC-02
5	REQ-05	System shall start a timer for the quiz.	N/A	DD-03	quiz.py	TC-05	STC-03
6	REQ-06	System shall record the user's selected answers.	ERD	DD-03	result.py	TC-06	STC-03
7	REQ-07	System shall calculate and display the final score.	ERD	DD-04	result.py	TC-07	STC-04
8	REQ-08	System shall display the leaderboard.	ERD	DD-04	leaderboard.py	TC-08	STC-04